Hennepin County SNAP/MFIP Report

Brandon Bloss, Ini Udomah, Vincent Rupp

Food insecurity remains a persistent issue in Hennepin County, despite the availability of federal assistance programs such as the Supplemental Nutrition Assistance Program (SNAP). While municipal-level averages often suggest stable or improving participation, these aggregated figures can mask important disparities at a more granular level, particularly within individual census tracts.This project investigates the extent to which county level averages obscure hyper-local areas with low SNAP participation. Specifically, we identify the most underrepresented census tracts across Hennepin County and analyze how participation in these areas has evolved from 2020 to 2025, with particular attention to changes following the end of pandemic era benefits in 2022. One key census tract we identified was around the dinky town U of M area where SNAP participation was low despite high numbers of people who qualify for SNAP benefits. Key stakeholders include county policymakers, public health and human service agencies, and nonprofit organizations working to address food insecurity. The populations most impacted by these findings are low-income households in underserved communities who may face barriers to accessing SNAP benefits. The purpose of this report is to provide a clearer, data driven understanding of participation gaps and to recommend more targeted, city level strategies to improve SNAP access and outreach.

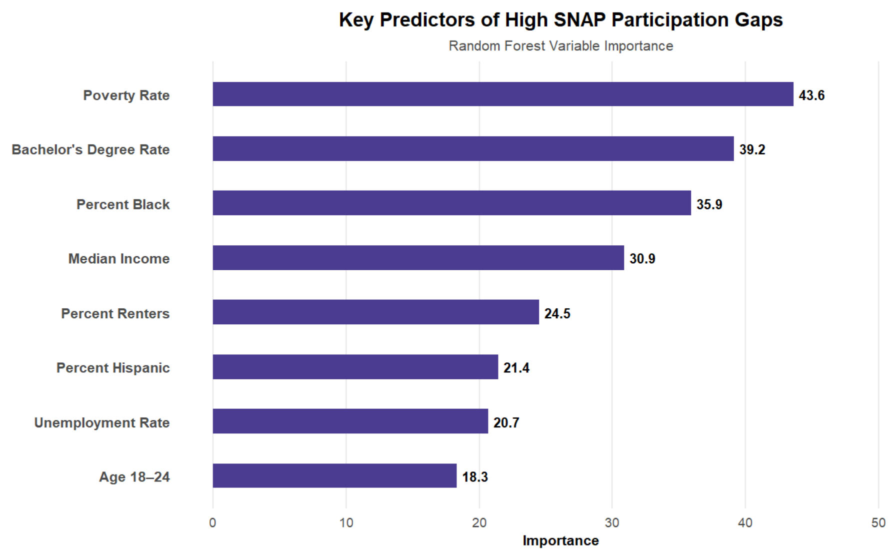
Our question was "To what extent do municipal boundaries hide hyper-local SNAP and MFIP participation hotspots and how has the density of this unmet need changed over time across Hennepin County post Covid from 2020 to 2025? Our project is designed to identify where Hennepin County residents are eligible to participate but are not enrolled, specifically looking for pockets that are "invisible" when the data is compiled at the city and county level.

Our primary data source is from Hennepin County itself, we have a dataset of SNAP/MFIP estimates at the census tract level that also include monthly enrollment counts. This also includes counts of residents living at or below 125% of the federally defined poverty line or FPL. The Hennepin County GeoJSON file of city borders and census tract boundaries from the U.S. Census Bureau were used to construct the map of Hennepin County visualization. The limitations that obstructed us were the vagueness of the 125% FPL because there are dynamic factors that determine acceptance in the program such as assets owned and household size. However, this varies significantly across various census tracts. Using a broader threshold than the standard 100%, we can see a more detailed account that can be pictured.

In Minnesota, SNAP and MFIP benefits are administered at the county level. The Human Services department of Hennepin County will use our findings to precisely target areas for outreach to maximize the impact of these services. Most reporting often uses city/county level averages. These averages tend to hide disparities that exist, especially in more affluent areas of the county. These areas within affluent cities are then disregarded and receive no attention. Our project is aiming to bypass the municipal boundaries to focus on finding the hidden hotspots for more effective resource allocation.

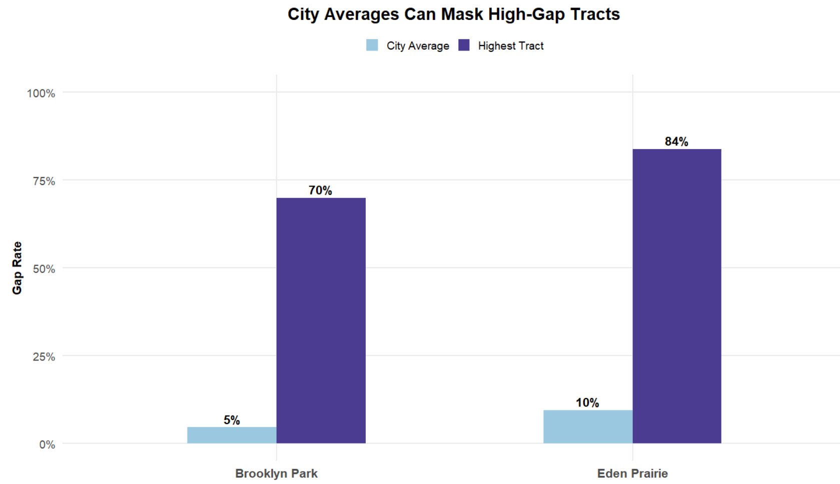
To identify the factors that are associated with high SNAP participation gaps, we built a classification model at the census tract level. After cleaning the SNAP/MFIP data, we formulated a 'gap rate', which is defined as: the difference between estimated eligible residents (≤ 125% FPL) and those enrolled in SNAP OR MFIP, divided by the eligible population. This formula is calculated at the census tract level. A 0.7 gap rate is easily interpreted as '70% of the estimated eligible population are NOT enrolled in SNAP or MFIP'. Tracts in the top 30% of gap rates were labeled "high gap". We combined this data with ACS data at the tract level, and included variables like income, race, education, age, employment, and housing. When calculating a yearly gap rate, the median of all months within the year was taken to represent the yearly gap rate. For ‘all years’ aggregated data, the median across all months (2020-2025) was taken.

We tested a decision tree and random forest model for this task, and we used an 80/20 test to train split. The decision tree was easier to interpret, but it was less accurate, while the random forest had a better overall performance with 83.3% accuracy, 71.4% sensitivity, and 84.7% specificity. While the model was strong at identifying low gap tract rates, it was also reasonably effective at identifying high gap tracts. We used the random forest as the primary model for evaluating which variables were most impactful in predicting high gap rate census tracts. The results showed poverty rate as the strongest predictor, followed by bachelor’s degree attainment, percent Black population, and median income. These variables should be interpreted as patterns and indicators of broader structural patterns, rather than direct causes of low SNAP/MFIP participation.



The results from our map reinforced the idea that participation gaps can be highly localized, and can vary vastly between even nearby geographic areas. To point out a few specific examples, there are several tracts in the University of Minnesota area that showcase extremely high gap rates despite large eligible populations. Census tract 38.02 has a gap rate of 92.7% (162 enrolled out of 2,234 eligible), while Census Tracts 1039 and 1049.01 both had gap rates of at least 97%, with over 1,000 estimated eligible residents each. These tracts are our best examples of concentrated areas of unmet SNAP/MFIP needs, and are areas of high opportunity for increasing enrollment.

More broadly speaking, the tract-level differences showcase how city-level averages can obscure more localized disparities. For example, Eden Prairie has a city-wide gap rate of 9.5%, but includes tracts with gap rates as high as 83.9%. Similarly, Brooklyn Park has a city-wide gap rate of 4.6%, but has tracts with gap rates as high as 70%. These examples are the focal point of our analysis. We have discovered instances like the above in virtually every municipality in Hennepin County that contains multiple census tracts. We believe that when discussing SNAP enrollment rates, numbers like 9.5% and 4.6% city-wide gap rates could lead to a lack of attention in those cities, despite there being neighborhoods in those cities that are experiencing low enrollment numbers.



It is important to note that we found 98 tracts (29.9%) have a gap rate of 0.0%. These values likely reflect data limitations, low count suppression, or possible mismatches between eligibility and observed enrollment. For the purposes of our project, we are focused on tracts with high gap rates, not low gap rates. With this in mind, we have elected to leave these 0.0% gap rates untouched. It is important to know and acknowledge that these 0% gap rates likely do not reflect reality, but we have left them as is, because we have reason to believe they are not tracts to be worried about in terms of SNAP/MFIP enrollment.

Overall, our analysis suggests that SNAP/MFIP enrollment gaps in Hennepin County are not evenly distributed and are often hidden when only looking at broad citywide averages. Instead, the data shows that disparities are concentrated in specific areas, with the largest gaps appearing in western, more rural parts of the county and around the University of Minnesota within the more urban eastern region. This pattern highlights that the issue is not just eligibility, but also awareness, accessibility, and local engagement. Many high-need census tracts are being overlooked because their challenges are masked by higher enrollment rates in surrounding areas. Based on these findings, we recommend that Hennepin County shift toward a more targeted, community level strategy. This includes partnering with the University of Minnesota to increase awareness among students through service booths at campus events, outreach in student housing, and social media campaigns tailored to that population. Additionally, expanding EBT acceptance by collaborating with local businesses near campus can reduce access barriers. In western Hennepin County, the county should work more closely with rural communities to address logistical challenges such as transportation, awareness, and application support. By focusing efforts at the census tract level and forming partnerships with local institutions, Hennepin County can more effectively close enrollment gaps and ensure that eligible residents are receiving the benefits they need.